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# A coupled elastic-plastic damage model for the mechanical behavior of three-dimensional (3D) braided composites

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ABSTRACT

A coupled elastic-plastic damage model is developed to describe the non-linear mechanical behavior of three-dimensional (3D) braided composites. In this model, the fiber breakage, inter-fiber fracture and matrix fracture are considered in the level of the fiber bundle and matrix. The onset and propagation of fiber bundle failure mechanisms are elastic and brittle, which is accounted for elastic damage model, and the elastic-plastic damage model used to describe the degradation of matrix is non-linear with progressive damage and inelastic strains. A set of internal variables are introduced to characterize the damage states of the fiber bundle and matrix and as a subsequence the degradation of the material stiffness. The damage initiation and propagation criteria are based on the Hashin criteria for the fiber bundle and the von Mises yield criterion for the matrix. The proposed damage model is implemented in the non-linear finite element analysis code ABAQUS using a user-subroutine UMAT to determine the response behavior of 3D braided composites under quasi-static loading, and the numerical predictions are compared with experimental data. The results predicted by the proposed model agree well with the experiment.



## 1. Introduction

The three-dimensional (3D) braided composites have a wide application in aerospace and other fields because of their excellent mechanical properties [1,2]. Thus an accurate evaluation of the mechanical characteristics becomes very important. The meso-structure and stiffness properties of the 3D braided composites have been extensively studied in recent decades [3–7], and the satisfactory results have now been obtained. However, due to the complexity of meso-structure and the diversity of damage modes for constituent materials, it is difficult to reveal the damage mechanisms from the results of mechanical tests alone, and the numerical investigations on the damage and failure of 3D braided composites have not been sufficiently clarified yet. Therefore, there is a need to conduct reliable simulations and analytical evaluations for damage behavior of 3D braided composites.

It is known that after the resin is infused and the composite is cured, some internal defects (such as micro-cracks, voids, etc.) in material may be present. These defects of material make it more susceptible to damage when subjected to the external loading. An accurate damage model to degrade the mechanical properties of composites is necessary. Besides, because of the stress concentration in the vicinity of the defects, the constituent materials also undergo some plastic deformations during loading so that the damage model cannot be applied alone. Therefore, the mechanical behavior of 3D braided composites is determined by two factors: damage and plasticity. Many researchers have used the damage theory alone to characterize the degradation of material stiffness by a set of damage variables which act on the elastic behavior [8–11]. McGregor et al. [8] combined the experimental and numerical study to calibrate the parameters of a macro-mechanical damage model. Applying the Murakami-Ohno damage theory, Fang et al. [9] studied the non-linear response behavior of 3D braided composites and considered that the damage accumulation eventually led to the failure of the composites. Wehrkamp-Richter et al. [10] investigated the damage and failure of tri-axial braided composites under multi-

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axial stress states and conducted uni-axial tensile tests at different off-axis orientations. Ivanov et al. [11] discussed the damage and failure behavior of tri-axial braided composites under tensile loading and the Puck's criterion was applied for damage initiation of fiber bundle. However, these works failed to evaluate the effect of plasticity on non-linear response of matrix material which has been observed by SEM in Refs. [3,12]. On the other hand, some researchers have adopted plasticity theory alone to represent the mechanical behavior of braided composites [13,14]. Song et al. [14] studied the compressive damage behavior of tri-axial braided composites and taken into account the inelastic properties of the matrix. Fang et al. [13] analyzed the compressive mechanical properties of 3D braided composites using the J2 flow theory of plasticity for matrix. However, the stiffness degradation of matrix material due to micro-cracks was not considered in these works. It has to be noted that the non-linear mechanical behavior of material is controlled by both plasticity and damage, thus one cannot be considered without the other.

On the basis of the aforementioned argument, it is essential to establish a model that is able to consider the coupling between plastic deformations and damage. In this paper, a coupled elastic-plastic damage model is established to analyze the damage development of 3D braided composites and validated with the experimental data.

This research is organized as follows: the specific damage model including initiation and evolution of damage is proposed in Section 2. Section 3 describes the numerical implementation of UMAT. The experiments and the simulations based on the representative volume cell (RVC) are presented in Section 4. And then, in Section 5, the simulated results are compared to corresponding experimental data and the damage development process is analyzed. Finally, the conclusions are given in Section 6.

## 2. Coupled elastic-plastic damage model

The 3D braided composites have two constituents: the matrix and the fibers. The fibers are in the form of fiber bundle in braided composites. The fiber bundle and matrix are treated macroscopically as transversely isotropic and isotropic homogeneous bodies, respectively. In this paper, the damage models for fiber bundle and matrix are formulated within the framework of Continuum Damage Mechanics (CDM) using internal variables, and based on the assumption of small strains.

### 2.1. Damage variables and constitutive functions

After the initiation of damage, the progression of damage is characterized by material stiffness degradation. The damage variables as internal variables can be used to describe the damage process. In this paper, the damage tensor suggested by Matzenmiller et al. [15] is adopted to computer the stiffness degradation. Since the fiber bundle shows no significant plastic deformation before failure, the elastic damage model is developed to describe the damage of fiber bundle. The constitutive equation of the damaged fiber bundle is defined as follows:

$$\boldsymbol{\varepsilon}_f = \boldsymbol{S}_f\left(\boldsymbol{d}_f\right):\boldsymbol{\sigma}_f \tag{1}$$

where $\boldsymbol{\varepsilon}_f$ and $\boldsymbol{\sigma}_f$ are the strain tensor and nominal stress tensor of fiber bundle, and the $\boldsymbol{S}_f(\boldsymbol{d}_f)$ is damaged compliance matrix:

$$\boldsymbol{S}_f\left(\boldsymbol{d}_f\right) = \begin{bmatrix} \frac{1}{\left(1-d_{f,1}\right)E_{f,1}} & -\frac{\nu_{f,12}}{E_{f,1}} & -\frac{\nu_{f,13}}{E_{f,1}} & 0 & 0 & 0 \\ & \frac{1}{\left(1-d_{f,2}\right)E_{f,2}} & -\frac{\nu_{f,23}}{E_{f,2}} & 0 & 0 & 0 \\ & & \frac{1}{\left(1-d_{f,3}\right)E_{f,3}} & 0 & 0 & 0 \\ & \text{sym.} & & \frac{1}{\left(1-d_{f,4}\right)G_{f,12}} & 0 & 0 \\ & & & & \frac{1}{\left(1-d_{f,5}\right)G_{f,23}} & 0 \\ & & & & & \frac{1}{\left(1-d_{f,6}\right)G_{f,31}} \end{bmatrix} \tag{2}$$

where the subscript 1 denotes the longitudinal direction of the fiber bundle, and 2 and 3 denote the transverse directions. $E_{f,1}$, $E_{f,2}$, $E_{f,3}$, $G_{f,12}$, $G_{f,23}$ and $G_{f,31}$ are undamaged material moduli, and $\nu_{f,12}$, $\nu_{f,23}$ and $\nu_{f,13}$ are undamaged material Poisson's ratios. $d_{f,1}$ is the damage variable for fiber breakage, denoted by subscripts $t$ and $c$ for tension and compression, and $d_{f,2}$ and $d_{f,3}$ are the damage variables for inter-fiber fracture in the fiber bundle, denoted by subscripts $t$ and $c$ for tension and compression. $d_{f,4}$ and $d_{f,6}$ are the damage variables for fiber fracture and inter-fiber fracture, and $d_{f,5}$ is associated with inter-fiber fracture. In addition, it is postulated that the damage variables $d_{f,4}$, $d_{f,5}$ and $d_{f,6}$ are not independent, and can be expressed as a function of the remaining variables:

$$\begin{aligned} d_{f,4} &= 1-\left(1-d_{f,1}\right)\left(1-d_{f,2}\right) \\ d_{f,5} &= 1-\left(1-d_{f,2}\right)\left(1-d_{f,3}\right) \\ d_{f,6} &= 1-\left(1-d_{f,3}\right)\left(1-d_{f,1}\right) \end{aligned} \tag{3}$$

According to the hypothesis of strain equivalence, the effective stress tensor of fiber bundle is computed as:

$$\tilde{\boldsymbol{\sigma}}_f = \boldsymbol{S}_{f0}^{-1} : \boldsymbol{\varepsilon}_f \tag{4}$$

where $\boldsymbol{S}_{f0}$ is the undamaged compliance tensor obtained from Eq. (2) using $d_{f,1} = d_{f,2} = d_{f,3} = d_{f,4} = d_{f,5} = d_{f,6} = 0$.

The matrix is assumed to be isotropic homogeneous body. It has to be noted that for several specific damage modes, the matrix with micro-cracks is anisotropic. To simplify matter, this paper does not take into account the anisotropic damage of matrix, and an isotropic elastic-plastic damage model is established to approximate the ductility degradation process due to micro-cracks. As classically, the total strain tensor is decomposed into an elastic part $\boldsymbol{\varepsilon}_m^e$ and a plastic part $\boldsymbol{\varepsilon}_m^p$:

$$\boldsymbol{\varepsilon}_m = \boldsymbol{\varepsilon}_m^e + \boldsymbol{\varepsilon}_m^p \tag{5}$$

The constitutive equations of the damaged matrix and undamaged matrix are presented in the following form:

$$\boldsymbol{\varepsilon}_m^e = \boldsymbol{S}_m(d_m) : \boldsymbol{\sigma}_m, \qquad \tilde{\boldsymbol{\sigma}}_m = \boldsymbol{S}_{m0}^{-1} : \boldsymbol{\varepsilon}_m^e \tag{6}$$

where $\boldsymbol{\sigma}_m$ and $\tilde{\boldsymbol{\sigma}}_m$ are the nominal stress tensor and effective stress tensor of the matrix, and the $\boldsymbol{S}_m\,(d_m)$ is damaged compliance matrix:

$$\boldsymbol{S}_m(d_m) = \frac{1}{E_m}\begin{bmatrix} \frac{1}{1-d_m} & -\nu_m & -\nu_m & 0 & 0 & 0 \\ & \frac{1}{1-d_m} & -\nu_m & 0 & 0 & 0 \\ & & \frac{1}{1-d_m} & 0 & 0 & 0 \\ & \text{sym.} & & \frac{2(1+\nu_m)}{1-d_m} & 0 & 0 \\ & & & & \frac{2(1+\nu_m)}{1-d_m} & 0 \\ & & & & & \frac{2(1+\nu_m)}{1-d_m} \end{bmatrix} \tag{7}$$

where $E_m$ is undamaged material elasticity modulus, and $\nu_m$ is undamaged material Poisson's ratio. $d_m$ is the damage variable for matrix micro-cracks, denoted by subscripts $t$ and $c$ for tension and compression. $\boldsymbol{S}_{m0}$ is the undamaged compliance tensor obtained from Eq. (7) using $d_m = 0$.

In this model, the effective stress is used to couple the plasticity and damage effects in a single constitutive equation for matrix, because it can provide a simple way to separate the damage and plastic processes. The isotropic damage accounts for the softening response and the decrease in the elastic stiffness, while the hardening plasticity is responsible for the development of irreversible strains. In other words, the effect of plastic deformation which is driven by the effective stresses can be described independently from damage ones and vice versa. In the following, the von Mises yield function [16] defined in the effective stress space is adopted to describe the plastic loading of the matrix in combination with an associated flow rule and isotropic hardening law. Assuming von Mises behavior [16], the plastic yield function is expressed in terms of effective stress as follows:

$$\begin{aligned} &F(\tilde{\boldsymbol{\sigma}}_m, \bar{\varepsilon}_m^p) = f(\tilde{\boldsymbol{\sigma}}_m) - \tilde{\sigma}_s(\bar{\varepsilon}_m^p) = 0 \\ &f(\tilde{\boldsymbol{\sigma}}_m) = \bar{\sigma}_m = \frac{1}{\sqrt{2}}\Big[(\tilde{\sigma}_{m,11} - \tilde{\sigma}_{m,22})^2 + (\tilde{\sigma}_{m,22} - \tilde{\sigma}_{m,33})^2 \\ &\qquad + (\tilde{\sigma}_{m,33} - \tilde{\sigma}_{m,11})^2 + 6\left(\tilde{\sigma}_{m,12}^2 + \tilde{\sigma}_{m,23}^2 + \tilde{\sigma}_{m,31}^2\right)\Big]^{\frac{1}{2}} \end{aligned} \tag{8}$$

where $f$ is the plastic potential, and $\tilde{\sigma}_s$ is the plastic hardening stress which depends only on the equivalent plastic strain $\bar{\varepsilon}_m^p$ and can be obtained from the experimental hardening curve. $\bar{\sigma}_m$ is the von Mises equivalent stress. $\tilde{\sigma}_{m,11}, \tilde{\sigma}_{m,22}, \tilde{\sigma}_{m,33}$ and $\tilde{\sigma}_{m,12}, \tilde{\sigma}_{m,23}, \tilde{\sigma}_{m,31}$ are the normal and shear stresses in the effective stress space, respectively.

Assuming the associated plastic flow rule for the matrix, the plastic strain rate $\dot{\boldsymbol{\varepsilon}}_m^p$ takes the form:

$$\dot{\boldsymbol{\varepsilon}}_m^p = \dot{\lambda}\boldsymbol{r} = \dot{\lambda}\frac{\partial F(\tilde{\boldsymbol{\sigma}}_m, \bar{\varepsilon}_m^p)}{\partial \tilde{\boldsymbol{\sigma}}_m} \tag{9}$$

where $\boldsymbol{r}$ is the plastic flow direction, and $\lambda$ is the non-negative plastic multiplier. The dot denotes time derivatives. On the basis of the equivalence of plastic work, the rate of the equivalent plastic strain is obtained:

$$\dot{\bar{\varepsilon}}_m^p = \left(\frac{2}{3}\dot{\boldsymbol{\varepsilon}}_m^p : \dot{\boldsymbol{\varepsilon}}_m^p\right)^{\frac{1}{2}} = \dot{\lambda} \tag{10}$$

### 2.2. Damage initiation criteria

In order to predict the damage initiation and propagation in the fiber bundle and matrix and evaluate the effective stress state, the damage activation functions for fiber bundle and matrix are defined as:

$$\begin{aligned} F_{f,I} &= \phi_{f,I} - r_{f,I}(I = 1t, 1c, 2t, 2c, 3t, 3c) \\ F_{m,J} &= \phi_{m,J} - r_{m,J}(J = t, c) \end{aligned} \tag{11}$$

where $\phi_{f,I}$ and $\phi_{m,J}$ are the loading functions for different failure modes of the fiber bundle and matrix which depend on the effective stress tensor and material constants. $r_{f,I}$ and $r_{m,J}$ are the damage threshold parameters, which take an initial vale of 1 when the material is undamaged and increase with the damage level. For the fiber bundle, the 3D Hashin failure criteria [17] is adopted, which is expressed in the effective stress space:

$$\phi_{f,1t} = \sqrt{\left(\frac{\tilde{\sigma}_{f,11}}{X_{f,1t}}\right)^2 + \left(\frac{\tilde{\sigma}_{f,12}}{S_{f,12}}\right)^2 + \left(\frac{\tilde{\sigma}_{f,31}}{S_{f,31}}\right)^2} \geq 1 \;\; if \;\; \tilde{\sigma}_{f,11} \geq 0$$

$$\phi_{f,1c} = \sqrt{\left(\frac{\tilde{\sigma}_{f,11}}{X_{f,1c}}\right)^2} \geq 1 \;\; if \;\; \tilde{\sigma}_{f,11} < 0$$

$$\phi_{f,2(3)t} = \sqrt{\left(\frac{\tilde{\sigma}_{f,22} + \tilde{\sigma}_{f,33}}{Y_{f,2t}}\right)^2 + \frac{\left(\tilde{\sigma}_{f,23}^2 - \tilde{\sigma}_{f,22}\tilde{\sigma}_{f,33}\right)}{S_{f,23}^2} + \left(\frac{\tilde{\sigma}_{f,12}}{S_{f,12}}\right)^2 + \left(\frac{\tilde{\sigma}_{f,31}}{S_{f,31}}\right)^2} \geq 1 \;\; if \;\; \tilde{\sigma}_{f,22} + \tilde{\sigma}_{f,33} \geq 0 \tag{12}$$

$$\phi_{f,2(3)c} = \sqrt{\frac{1}{Y_{f,2c}}\left[\left(\frac{Y_{f,2c}}{2S_{f,23}}\right)^2 - 1\right]\left(\tilde{\sigma}_{f,22} + \tilde{\sigma}_{f,33}\right) + \left(\frac{\tilde{\sigma}_{f,22} + \tilde{\sigma}_{f,33}}{2S_{f,23}}\right)^2 + \frac{\left(\tilde{\sigma}_{f,23}^2 - \tilde{\sigma}_{f,22}\tilde{\sigma}_{f,33}\right)}{S_{f,23}^2} + \left(\frac{\tilde{\sigma}_{f,12}}{S_{f,12}}\right)^2 + \left(\frac{\tilde{\sigma}_{f,31}}{S_{f,31}}\right)^2} \geq 1$$

$$if \, \tilde{\sigma}_{f,22} + \tilde{\sigma}_{f,33} < 0$$

where $X_{f,1t}$ and $X_{f,1c}$ are the tensile and compressive strengths of fiber bundle in the fiber direction; $Y_{f,2t}$ and $Y_{f,2c}$ are the transverse tensile and compressive strengths of fiber bundle; $S_{f,12}$, $S_{f,23}$ and $S_{f,31}$ are the shear strengths of fiber bundle. It has to be mentioned that the Hashin criteria have been widely applied in the engineering, and it would be relatively straightforward to incorporate alternative initiation criteria.

For the matrix, the initiation criteria are based on the von Mises theory [16]:

$$\begin{aligned} \phi_{m,t} &= \frac{\overline{\sigma}_m}{X_{m,t}} \geq 1 \;\; \text{if} \;\; \tilde{I}_{m,1} = \tilde{\sigma}_{m,11} + \tilde{\sigma}_{m,22} + \tilde{\sigma}_{m,33} \geq 0 \\ \phi_{m,c} &= \frac{\overline{\sigma}_m}{X_{m,c}} \geq 1 \;\; \text{if} \;\; \tilde{I}_{m,1} = \tilde{\sigma}_{m,11} + \tilde{\sigma}_{m,22} + \tilde{\sigma}_{m,33} < 0 \end{aligned} \tag{13}$$

where $X_{m,t}$ and $X_{m,c}$ are the tensile and compressive strengths of matrix, respectively; $\tilde{I}_{m,1}$ is the first-invariant of the effective stress tensor of matrix.

### 2.3. Damage evolution

Once the damage initiation criteria are satisfied, further loading will cause degradation of material stiffness. The damage process of material is an irreversible deformation thermodynamically, thus the damage threshold values $r_N$ must satisfy the Kuhn-Tucker and consistency conditions:

$$\begin{aligned} &\dot{r}_N \geq 0; F_N \leq 0; \dot{r}_N F_N = 0 \\ &\dot{F}_N = \dot{\phi}_N - \dot{r}_N = 0 \end{aligned} \tag{14}$$

By integrating the equations, the expressions of the damage threshold factors $r_N$ can be obtained:

$$\begin{aligned} r_{f,I} &= \max\left\{1, \max\left\{\phi_{f,I}^{\tau}\right\}\right\} \;\; \tau \in [0,t], (I = 1t, 1c, 2t, 2c, 3t, 3c) \\ r_{m,J} &= \max\left\{1, \max\left\{\phi_{m,J}^{\tau}\right\}\right\} \;\; \tau \in [0,t], (J = t, c) \end{aligned} \tag{15}$$

where $t$ is the total time.

The fiber bundle generally displays the typical elastic-brittle behavior and the exponential damage evolution laws [18] (see Fig. 1(a)) can be used to represent the stiffness degradation of fiber bundle but not for the fiber dominated longitudinal tensile damage denoted as $d_{f,1t}$. In order to simulate the phenomenon of the fiber bridging and fiber pull-out in the fiber bundle, the linear and exponential damage evolution laws [19,20] (see Fig. 1(b)) are applied for the fiber bundle in the longitudinal direction:

$$\begin{aligned} d_{f,I} &= 1 - \frac{1}{r_{f,I}}\exp\left[A_{f,I}\left(1 - r_{f,I}\right)\right] \;\; (I = 1c, 2t, 2c, 3t, 3c) \\ d_{f,1t} &= 1 - \frac{1 - d_{f,1t}^L}{r_{f,1t}^E}\exp\left[A_{f,1t}\left(1 - r_{f,1t}^E\right)\right] \end{aligned} \tag{16}$$

where $A_{f,I}$ is the exponential damage softening factor, and $d_{f,1t}^L$, $r_{f,1t}^E$ are the auxiliary damage variables:

$$\begin{aligned} d_{f,1t}^L &= 1 + \frac{K_{f,1}}{E_{f,1}} - \left(1 + \frac{K_{f,1}}{E_{f,1}}\right)\frac{1}{r_{f,1t}^L} \\ r_{f,1t}^L &= \max\left\{1, \min\left(r_{f,1t}, r_{f,1t}^F\right)\right\} \\ r_{f,1t}^E &= \max\left\{1, \left(1 - d_{f,1t}^F\right)\frac{X_{f,1t}}{X_{f,PO}} r_{f,1t}\right\} \end{aligned} \tag{17}$$

where $K_{f,1}$ is the slope of the linear softening law, and $r_{f,1t}^L$ is the auxiliary threshold value. $r_{f,1t}^F$ is the damage threshold value at the transition between the linear and exponential damage evolution laws, and $d_{f,1t}^F$, $X_{f,PO}$ are the corresponding values of the damage variable and stress, respectively.

The failure mode of the matrix is different from that of the fiber bundle. It can be seen from the experimental data in the literature that the resin matrix shows a certain degree of plasticity under both quasi-static tensile and compressive loads [3,14]. Therefore, as sketched in Fig. 2, the exponential damage evolution law [18] is adopted to characterize the elastic-plastic damage of the matrix:

$$d_{m,J} = 1 - \frac{1}{r_{m,J}}\exp\left[A_{m,J}\left(1 - r_{m,J}\right)\right] (J = t, c) \tag{18}$$

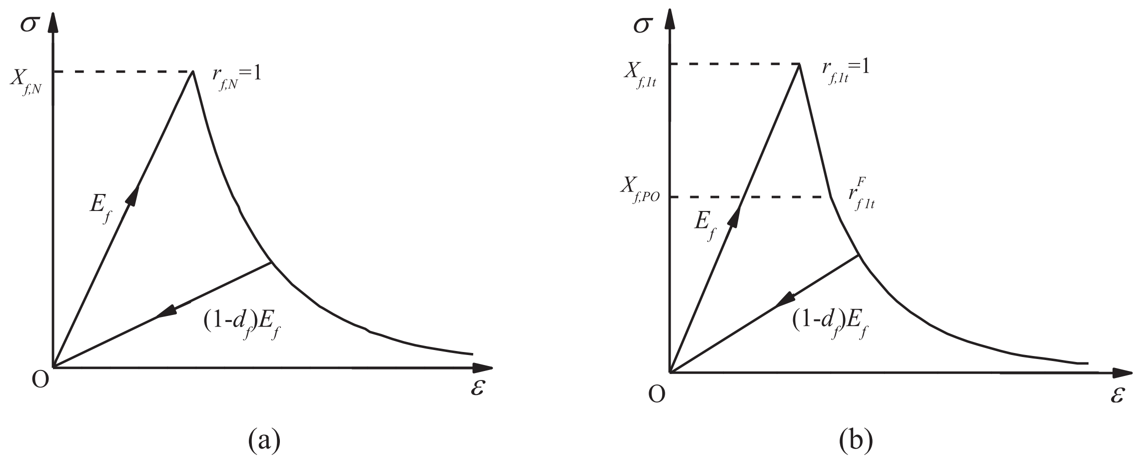


**Fig. 1.** Damage evolution models of the fiber bundle: (a) the exponential damage evolution law and (b) the linear and exponential damage evolution law.

where $A_{mJ}$ is a parameter that defines the exponential softening law of the matrix.

In the finite element calculation for the damage model, the numerical simulation results are significantly dependent on the finite element mesh size. In order to avoid the influence of the finite element mesh size on the calculation results, all the softening parameters $A_M$ must be obtained by regularizing the strain-softening portion of the stress-strain curve, which is based on the crack band theory proposed by Bazant [21]. According to this theory, the energy dissipated per unit volume for uniaxial condition $g_M$ can be determined by the fracture toughness $G_M$ and characteristic length of the finite element$l^*$:

$$g_M = \frac{G_M}{l^*}(M = f,1t; f,1c; f,2t; f,2c; f,3t; f,3c; m,t; m,c) \tag{19}$$

The energy dissipated per unit volume for uniaxial stress conditions can be determined by integrating the damage energy dissipation rate:

$$g_M = \int_0^{\infty} \frac{\partial G}{\partial d_M}\dot{d}_M dt = \int_1^{\infty} \frac{\partial G}{\partial d_M}\frac{\partial d_M}{\partial r_M} dr_M \tag{20}$$

where $G$ is the Helmholtz free energy. Combining Eqs. (19) and (20) yields the following expression:

$$\int_1^{\infty} \frac{\partial G}{\partial d_M}\frac{\partial d_M}{\partial r_M} dr_M = \frac{G_M}{l^*} \tag{21}$$

The softening parameters $A_M$ obtained by Eq. (21) can ensure that the calculated dissipated energy is independent of mesh refinement. For fiber bundle, the Helmholtz free energy expression can be written as:

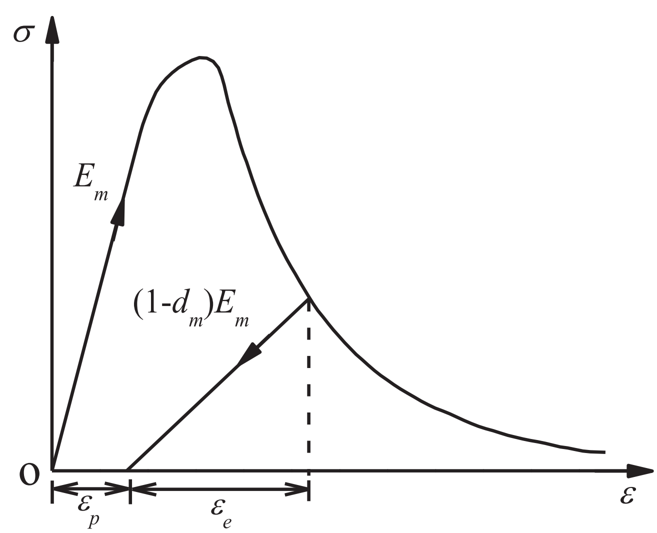


**Fig. 2.** Damage evolution law of the matrix.

For matrix, the Helmholtz free energy $G_m$ is the sum of the thermo-elastic $G_m^e$ and the plastic $G_m^p$ contributions. The elastic contribution $G_m^e$ is affected by damage to model the experimentally-observed coupling between elasticity and damage through the effective stress, and the plastic part $G_m^p$ is the contribution due to plastic hardening [16]:

$$G_m = G_m^e(\boldsymbol{\varepsilon}_m^e, d_m) + G_m^p(\bar{\varepsilon}_m^p) \tag{23}$$

Eq. (23) can be rewritten as:

$$G_f = \frac{1}{2}\boldsymbol{\sigma}_f:\boldsymbol{\varepsilon}_f = \frac{\sigma_{f,11}^2}{2\left(1-d_{f,1}\right)E_{f,1}} + \frac{\sigma_{f,22}^2}{2\left(1-d_{f,2}\right)E_{f,2}} + \frac{\sigma_{f,33}^2}{2\left(1-d_{f,3}\right)E_{f,3}} - \frac{\nu_{f,12}}{E_{f,1}}\sigma_{f,11}\sigma_{f,22} - \frac{\nu_{f,13}}{E_{f,1}}\sigma_{f,11}\sigma_{f,33}$$
$$-\frac{\nu_{f,23}}{E_{f,2}}\sigma_{f,22}\sigma_{f,33} + \frac{\sigma_{f,12}^2}{2\left(1-d_{f,4}\right)G_{f,12}} + \frac{\sigma_{f,23}^2}{2\left(1-d_{f,5}\right)G_{f,23}} + \frac{\sigma_{f,31}^2}{2\left(1-d_{f,6}\right)G_{f,31}} \tag{22}$$

$$G_m = \frac{\sigma_{m,11}^2 + \sigma_{m,22}^2 + \sigma_{m,33}^2}{2(1-d_m)E_m} - \frac{\nu_m}{E_m}(\sigma_{m,11}\sigma_{m,22} + \sigma_{m,22}\sigma_{m,33} + \sigma_{m,33}\sigma_{m,11}) + \frac{(1+\nu_m)\left(\sigma_{m,12}^2 + \sigma_{m,23}^2 + \sigma_{m,31}^2\right)}{(1-d_m)E_m} + G_m^p(\bar{\varepsilon}_m^p) \tag{24}$$

Then, the softening parameters $A_M$ can be determined by solving the numerical integration equation Eq. (21) using the integration algorithm presented in Supplementary Appendix A.

## 3. Numerical implementation

The proposed constitutive model is implemented in the non-linear finite element analysis code ABAQUS [22] using a user-subroutine UMAT, which is based on the classic step by step iterative method in an incremental form.

### 3.1. Stress update algorithm for plastic process

The plastic part for the matrix in Eq. (5) is path dependent and requires an iterative process to be solved. These relations are integrated with the backward Euler implicit integration scheme (between loading steps $t_n$ and $t_{n+1}$) according to a closest point projection [23]. The scheme consists of an initial elastic-predictor step, involving an excursion (in effective stress space) away from the yield surface, and a plastic-corrector step which returns the stress to the updated yield surface.

It yields the non-linear system of equations in the effective stress space based on incremental approach:

$$\begin{aligned}
&\boldsymbol{\varepsilon}_{m,n+1} = \boldsymbol{\varepsilon}_{m,n} + \Delta\boldsymbol{\varepsilon}_{m,n+1}\\
&\boldsymbol{\varepsilon}_{m,n+1}^p = \boldsymbol{\varepsilon}_{m,n}^p + \Delta\boldsymbol{\varepsilon}_{m,n+1}^p = \boldsymbol{\varepsilon}_{m,n}^p + \Delta\lambda_{n+1}\boldsymbol{r}_{n+1}\\
&\bar{\varepsilon}_{m,n+1}^p = \bar{\varepsilon}_{m,n}^p + \Delta\bar{\varepsilon}_{m,n+1}^p = \bar{\varepsilon}_{m,n}^p + \Delta\lambda_{n+1}\\
&\tilde{\boldsymbol{\sigma}}_{m,n+1} = \left(\tilde{\boldsymbol{\sigma}}_{m,n} + \boldsymbol{S}_{m0}^{-1} : \Delta\boldsymbol{\varepsilon}_{m,n+1}\right) - \boldsymbol{S}_{m0}^{-1} : \Delta\boldsymbol{\varepsilon}_{m,n+1}^p = \tilde{\boldsymbol{\sigma}}_{m,n+1}^{trial} - \boldsymbol{S}_{m0}^{-1} : \Delta\boldsymbol{\varepsilon}_{m,n+1}^p\\
&F_{n+1} = F\left(\tilde{\boldsymbol{\sigma}}_{m,n+1}, \bar{\varepsilon}_{m,n+1}^p\right) = f(\tilde{\boldsymbol{\sigma}}_{m,n+1}) - \tilde{\sigma}_{s,n+1}\left(\bar{\varepsilon}_{m,n+1}^p\right) = 0
\end{aligned} \tag{25}$$

where $\boldsymbol{\varepsilon}_{m,n}^p$, $\bar{\varepsilon}_{m,n}^p$ and the total strains $\boldsymbol{\varepsilon}_{m,n+1}$ are known at steps $t_n$ and $t_{n+1}$, respectively. The unknowns of the local problem are $\tilde{\boldsymbol{\sigma}}_{m,n+1}$, $\bar{\varepsilon}_{m,n+1}^p$ and the incremental multiplier denoted as $\Delta\lambda_{n+1}$ in this section. The solution of the non-linear system of equations is typically obtained by an iterative Newton-Raphson method. Note that the following initial values are

$$\begin{aligned}
\boldsymbol{\varepsilon}_{m,n+1}^{p,(0)} &= \boldsymbol{\varepsilon}_{m,n}^p \bar{\varepsilon}_{m,n+1}^{p,(0)} = \bar{\varepsilon}_{m,n}^p \Delta\lambda_{n+1}^{(0)} = 0\tilde{\boldsymbol{\sigma}}_{m,n+1}^{(0)} = \tilde{\boldsymbol{\sigma}}_{m,n+1}^{trial}\\
&= \tilde{\boldsymbol{\sigma}}_{m,n} + \boldsymbol{S}_{m0}^{-1} : \Delta\boldsymbol{\varepsilon}_{m,n+1}
\end{aligned} \tag{26}$$

$\delta\Delta\lambda_{n+1}^{(k)}$ is the increment in $\Delta\lambda_{n+1}$ at the $k$th iteration:

$$\begin{aligned}
&\delta\Delta\lambda_{n+1}^{(k)} = \frac{\left(\bar{\sigma}_{m,n+1}^{(0)} - 3\mu\Delta\lambda_{n+1}^{(k)}\right) - \tilde{\sigma}_{s,n+1}^{(k)}\left(\bar{\varepsilon}_{m,n+1}^{p,(k)}\right)}{3\mu + H_{n+1}^{(k)}}\\
&\bar{\sigma}_{m,n+1}^{(0)} = \frac{1}{\sqrt{2}}\left[\left(\tilde{\sigma}_{m,n+1,11}^{(0)} - \tilde{\sigma}_{m,n+1,22}^{(0)}\right)^2 + \left(\tilde{\sigma}_{m,n+1,22}^{(0)} - \tilde{\sigma}_{m,n+1,33}^{(0)}\right)^2 + \left(\tilde{\sigma}_{m,n+1,33}^{(0)} - \tilde{\sigma}_{m,n+1,11}^{(0)}\right)^2 + 6\left(\tilde{\sigma}_{m,n+1,12}^{(0)} + \tilde{\sigma}_{m,n+1,23}^{(0)} + \tilde{\sigma}_{m,n+1,31}^{(0)}\right)\right]^{\frac{1}{2}}\\
&H_{n+1}^{(k)} = \frac{d\tilde{\sigma}_{s,n+1}^{(k)}}{d\bar{\varepsilon}_{m,n+1}^{p,(k)}}
\end{aligned} \tag{27}$$

where $\bar{\sigma}_{m,n+1}^{(0)}$ is the initial von Mises equivalent stress of matrix, and $\mu$ is the shear modulus of matrix. Thus, the updated values of the incremental multiplier, equivalent plastic strain, plastic strain and effective stress with the increment as given in Eq. (27) are

$$\begin{aligned}
\Delta\lambda_{n+1}^{(k+1)} &= \Delta\lambda_{n+1}^{(k)} + \delta\Delta\lambda_{n+1}^{(k)}\\
\bar{\varepsilon}_{m,n+1}^{p,(k+1)} &= \bar{\varepsilon}_{m,n}^p + \Delta\lambda_{n+1}^{(k+1)}\\
\boldsymbol{\varepsilon}_{m,n+1}^{p,(k+1)} &= \boldsymbol{\varepsilon}_{m,n}^p + \Delta\lambda_{n+1}^{(k+1)}\boldsymbol{r}_{n+1}\\
\tilde{\boldsymbol{\sigma}}_{m,n+1}^{(k+1)} &= \left(\tilde{\boldsymbol{\sigma}}_{m,n+1}^{trial}\right) - \Delta\lambda_{n+1}^{(k+1)}\boldsymbol{S}_{m0}^{-1}:\boldsymbol{r}_{n+1}
\end{aligned} \tag{28}$$

The Newton-Raphson procedure is continued until the convergence to the updated yield surface is achieved to within a sufficient tolerance. It is noted that the variables are updated from the converged values at the end of the previous time-step.

### 3.2. Viscous regularization for damage variables

Strain-softening constitutive models and stiffness degradation often lead to severe convergence difficulties in implicit analysis programs. Some of these convergence difficulties can be alleviated by using a viscous regularization of the constitutive equations. The Duvaut-Lions regularization model [24] is applied:

$$\dot{d}^{v}_{f,I} = \frac{1}{\eta}\left(d_{f,I} - d^{v}_{f,I}\right)(I = 1,2,3)$$
$$\dot{d}^{v}_{m} = \frac{1}{\eta}\left(d_{m} - d^{v}_{m}\right) \tag{29}$$

where $\eta$ is the viscous coefficient which represents the relaxation time of the viscous system, and $d^{v}_{f,I}$, $d^{v}_{m}$ denote regularized damage variables. The response behavior of the damaged material is calculated using the regularized damage variables. A backward-Euler scheme is used to evaluate the time integration of the damage variables in the finite element method:

$$d^{v}_{f,I,n+1} = \frac{\Delta t}{\eta + \Delta t} d_{f,I,n+1} + \frac{\eta}{\eta + \Delta t} d^{v}_{f,I,n}(I = 1,2,3)$$
$$d^{v}_{m,n+1} = \frac{\Delta t}{\eta + \Delta t} d_{m,n+1} + \frac{\eta}{\eta + \Delta t} d^{v}_{m,n} \tag{30}$$

Note that, it is important to use the viscosity parameter with a small value (small compared to the characteristic time increment) which can help improving the rate of convergence of the model in the softening regime, without compromising results.

### 3.3. Material tangent constitutive tensor and computational procedure

In order to obtain a fast convergence rate of the solution algorithm for the non-linear system of equations, the consistent tangent operator for the proposed constitutive model of fiber bundle is derived in the following form:

$$\boldsymbol{C}_{f,t} = \frac{\partial \boldsymbol{\sigma}_{f,n+1}}{\partial \boldsymbol{\varepsilon}_{f,n+1}} = \boldsymbol{S}_{f}^{-1}\left(\boldsymbol{d}^{v}_{f,n+1}\right):\left[\boldsymbol{I} - \boldsymbol{M}_{f}\left(\boldsymbol{d}^{v}_{f,n+1}\right)\right]$$
$$\boldsymbol{M}_{f}\left(\boldsymbol{d}^{v}_{f,n+1}\right) = \sum_{I} \frac{\partial \boldsymbol{S}_{f}\left(\boldsymbol{d}^{v}_{f,n+1}\right)}{\partial d^{v}_{f,I,n+1},} : \boldsymbol{\sigma}_{f,n+1} \frac{\Delta t}{\eta + \Delta t} \frac{\partial d_{f,I,n+1}}{\partial \boldsymbol{\varepsilon}_{f,n+1}}(I = 1,2,3) \tag{31}$$

where $\boldsymbol{I}$ is the identity tensor. The components of tensor $\boldsymbol{M}_f$ are presented in Supplementary Appendix B.

At each loading step, an increment of total strains is prescribed to each Gauss point. The corresponding increments of stress and damage variables are calculated by using the constitutive equations. The general computation procedure for the $(n+1)$th step can be summarized as follows:

(1). Read the strain tensor at the (n+1)th step: $\boldsymbol{\varepsilon}_{f,n+1} = \boldsymbol{\varepsilon}_{f,n} + \Delta\boldsymbol{\varepsilon}_{f,n+1}$
(2). Compute the effective stress tensor: $\tilde{\boldsymbol{\sigma}}_{f,n+1} = \boldsymbol{S}_{f0}^{-1}:\boldsymbol{\varepsilon}_{f,n+1}$
(3). Compute the loading functions: $\phi_{f,I,n+1}(\tilde{\boldsymbol{\sigma}}_{f,n+1})$
(4). Compute the threshold values: $r_{f,I,n+1}(r_{f,I,n}, \phi_{f,I,n+1})$
(5). Compute the damage variables: $d_{f,I,n+1}(r_{f,I,n+1}), d^{v}_{f,I,n+1}(d_{f,I,n+1}, d^{v}_{f,I,n})$
(6). Compute the nominal stress tensor: $\boldsymbol{\sigma}_{f,n+1} = \boldsymbol{S}_{f}^{-1}(\boldsymbol{d}^{v}_{f,n+1}):\boldsymbol{\varepsilon}_{f,n+1}$
(7). Compute the consistent tangent tensor: $\boldsymbol{C}_{f,t} = \frac{\partial \boldsymbol{\sigma}_{f,n+1}}{\partial \boldsymbol{\varepsilon}_{f,n+1}}$

To ensure the quadratic convergence rate of the Newton-Raphson method for the elastic-plastic damage of matrix, the consistent tangent operator for the presented constitutive model of matrix is calculated in the following form:

$$\boldsymbol{C}_{m,t} = \frac{\partial \boldsymbol{\sigma}_{m,n+1}}{\partial \boldsymbol{\varepsilon}_{m,n+1}} = \boldsymbol{S}_{m}^{-1}\left(d^{v}_{m,n+1}\right):\left(\boldsymbol{I} - \boldsymbol{M}_{m}\left(d^{v}_{m,n+1}\right)\right):\boldsymbol{S}_{m0}^{-1}:\frac{\partial \tilde{\boldsymbol{\sigma}}_{m,n+1}}{\partial \boldsymbol{\varepsilon}_{m,n+1}}$$
$$\boldsymbol{M}_{m}\left(d^{v}_{m,n+1}\right) = \frac{\partial \boldsymbol{S}_{m}\left(d^{v}_{m,n+1}\right)}{\partial d^{v}_{m,n+1}} : \boldsymbol{\sigma}_{n+1} \frac{\Delta t}{\eta + \Delta t} \frac{\partial d_{m,n+1}}{\partial \boldsymbol{\varepsilon}^{e}_{m,n+1}}$$
$$\frac{\partial \tilde{\boldsymbol{\sigma}}_{m,n+1}}{\partial \boldsymbol{\varepsilon}_{m,n+1}} = \tilde{\boldsymbol{C}}_{m,n+1} - \frac{\left(\tilde{\boldsymbol{C}}_{m,n+1}:\frac{\partial F_{n+1}}{\partial \tilde{\boldsymbol{\sigma}}_{n+1}}\right) \otimes \left(\tilde{\boldsymbol{C}}_{m,n+1}:\frac{\partial F_{n+1}}{\partial \tilde{\boldsymbol{\sigma}}_{n+1}}\right)}{\left(\frac{\partial F_{n+1}}{\partial \tilde{\boldsymbol{\sigma}}_{n+1}}\right):\tilde{\boldsymbol{C}}_{m,n+1}:\frac{\partial F_{n+1}}{\partial \tilde{\boldsymbol{\sigma}}_{n+1}} + \frac{d\tilde{\sigma}_{m,n+1}}{d\bar{\varepsilon}_{m,n+1}}} \tag{32}$$

where $\tilde{\boldsymbol{C}}_{m,n+1}$ is an equivalent stiffness matrix:

$$\tilde{\boldsymbol{C}}_{m,n+1} = \left(\boldsymbol{S}_{m0} + \Delta\lambda_{n+1}\frac{\partial^2 F_{n+1}}{\partial \tilde{\boldsymbol{\sigma}}^2_{m,n+1}}\right)^{-1} \tag{33}$$

The components of tensor $\boldsymbol{M}_m$ are presented in Supplementary Appendix B.

The corresponding increments of the stress, plastic strains and

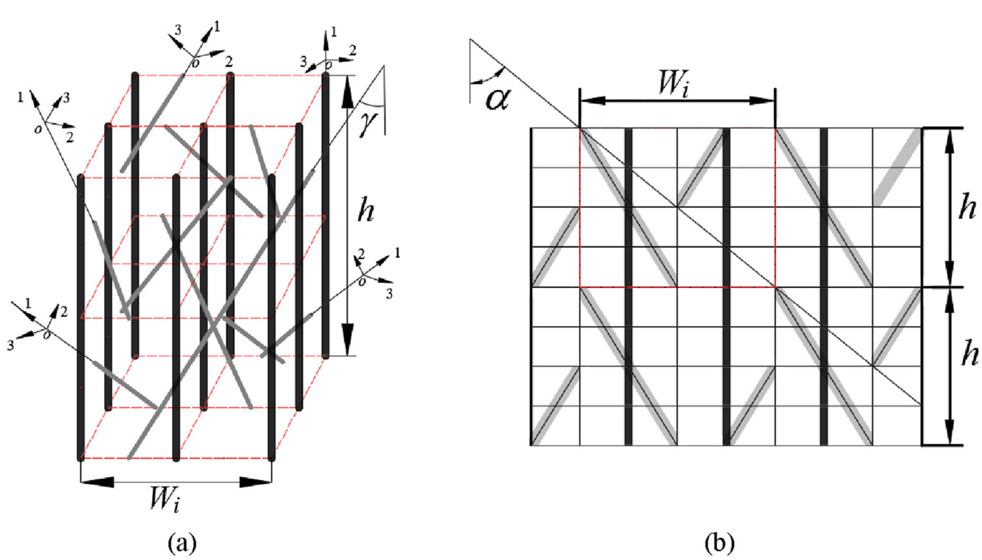


**Fig. 3.** The topological relation and preform surface diagram: (a) topological relationship and (b) preform surface diagram [4].

**Table 1**
Braided parameters and RVC geometry parameters [25].

| Specimen | Braided angle $\alpha$ (°) | Fiber volume fraction $V_f$ (%) | Width of RVC $W_i$ (mm) | Height of RVC $h$ (mm) |
|---|---|---|---|---|
| 1 | 20 | 45 | 3.052 | 8.415 |
| 2 | 40 | 50 | 3.357 | 4.012 |

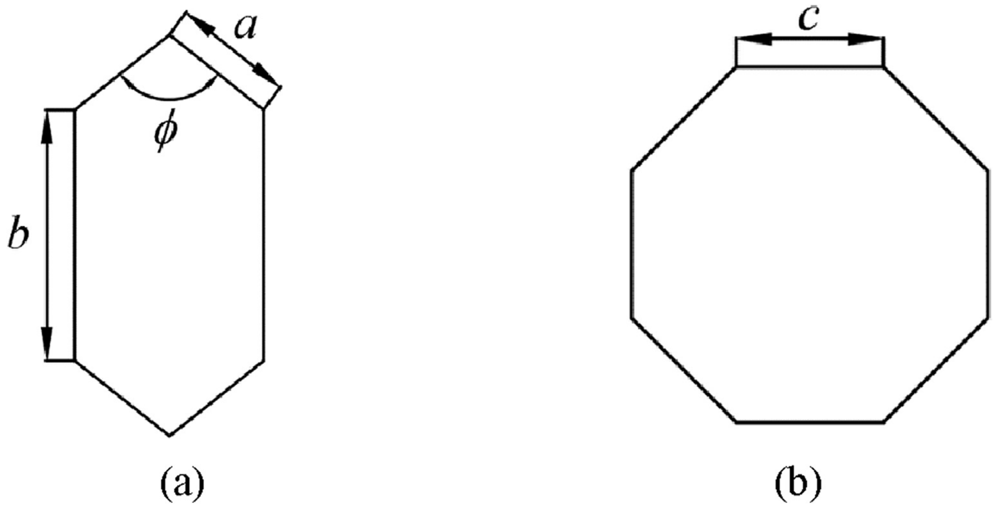


**Fig. 4.** The cross section of braiding yarn and axial yarn: (a) the braiding yarn and (b) the axial yarn.

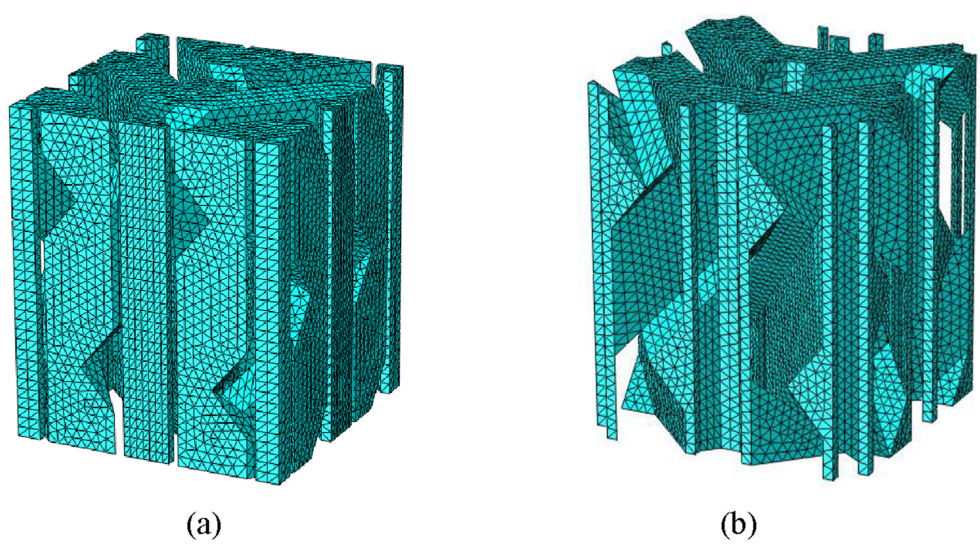


**Fig. 5.** Finite element model of constituent materials: (a) yarns and (b) resin matrix.

damage variable are calculated by using the constitutive equations. The computational procedure for the constitutive model of matrix is performed as follows:

(1). Read the variables at the $n$th step and the strain increment tensor at the $(n+1)$th step: $\boldsymbol{\varepsilon}_{m,n}, \boldsymbol{\varepsilon}^p_{m,n}, \bar{\varepsilon}^p_{m,n}, \tilde{\boldsymbol{\sigma}}_{m,n}, \boldsymbol{\sigma}_{m,n}, r_{m,n}, d_{m,n}, d^v_{m,n}, \Delta\boldsymbol{\varepsilon}_{m,n+1}, \boldsymbol{\varepsilon}_{m,n+1} = \boldsymbol{\varepsilon}_{m,n} + \Delta\boldsymbol{\varepsilon}_{m,n+1}$
(2). Compute the effective stress tensor using the backward Euler implicit integration scheme: $\tilde{\boldsymbol{\sigma}}_{m,n+1} = \mathbf{S}^{-1}_{m0}:(\boldsymbol{\varepsilon}_{m,n+1} - \boldsymbol{\varepsilon}^p_{m,n+1})$
(3). Compute the loading functions: $\phi_{m,n+1}(\tilde{\boldsymbol{\sigma}}_{m,n+1})$
(4). Compute the threshold values: $r_{m,n+1}(r_{m,n}, \phi_{m,n+1})$
(5). Compute the damage variables: $d_{m,n+1}(r_{m,n+1}), d^v_{m,n+1}(d_{m,n+1}, d^v_{m,n})$

**Table 2**
Material properties of T300 [25].

| | $E_1$ (GPa) | $E_2$ (GPa) | $G_{12}$ (GPa) | $G_{23}$ (GPa) | $\mu_{12}$ | $X_t$ (MPa) | $X_c$ (MPa) |
|---|---|---|---|---|---|---|---|
| T300 | 230 | 40 | 24 | 14.3 | 0.26 | 3580 | 2470 |

(6). Compute the nominal stress tensor: $\boldsymbol{\sigma}_{m,n+1} = \mathbf{S}^{-1}_m(d^v_{m,n+1}):(\boldsymbol{\varepsilon}_{m,n+1} - \boldsymbol{\varepsilon}^p_{m,n+1})$
(7). Compute the consistent tangent tensor: $\mathbf{C}_{m,t} = \frac{\partial\boldsymbol{\sigma}_{m,n+1}}{\partial\boldsymbol{\varepsilon}_{m,n+1}}$

## 4. Application to the 3D braided composites

The coupled elastic-plastic damage model is applied to

**Table 3**
Material properties of matrix and yarn [27,29].

| | | | | | | |
|---|---|---|---|---|---|---|
| Matrix | $E_m$ (GPa) 3.2 | $\nu_m$ 0.35 | $X_{m,t}$ (MPa) 75 | $X_{m,c}$ (MPa) 180 | $S_{m,s}$ (MPa) 60 | $G_{m,t(c)}$ (N/mm) 1.0 |
| Yarn | $E_{f,1}$ (GPa) 196.03 | $E_{f,2(3)}$ (GPa) 22.05 | $\nu_{f,12(13)}$ 0.268 | $\nu_{f,23}$ 0.375 | $G_{f,12(13)}$ (GPa) 9.35 | $G_{f,23}$ (GPa) 7.38 |
| | $X_{f,1t}$ (MPa) 2556 | $X_{f,1c}$ (MPa) 1263 | $Y_{f,2t}$ (MPa) 70 | $Y_{f,2c}$ (MPa) 170 | $S_{f,12(13)}$ (MPa) 58 | $S_{f,23}$ (MPa) 46 |
| | $G_{f,1t}$ (N/mm) 12.5 | $G_{f,1c}$ (N/mm) 12.5 | $G_{f,2(3)t}$ (N/mm) 1.0 | $G_{f,2(3)c}$ (N/mm) 1.0 | | |

investigate the non-linear mechanical behavior of 3D five-directional braided composites. The results of the numerical simulations are compared to corresponding experimental data.

### 4.1. Quasi-static tensile experiments

To validate the proposed model, the quasi-static tensile experiments of 3D five-directional braided composites taken from Ref. [12] were considered. The specimens of carbon/phenolic composites with the two different braiding angles (one is 20°, and the other is 40°) were tested. The axial yarns and braiding yarns were T300 carbon 6K and 9K, respectively. The phenolic resin was used for matrix. All the considered specimens were tested at room temperature. Additional details on the experimental technique and the test results can be found in Ref. [12].

### 4.2. RVC model

The topological relationship of main yarns in the RVC and preform surface of 3D five-directional braided composites are shown in Fig. 3(a) and Fig. 3(b), respectively [4]. The braid angle $\alpha$, pitch width $W_i$ and pitch length $h$ can be obtained by measuring the specimen. The braided parameters and RVC geometry parameters taken from Ref. [25] are given in Table 1. Based on the geometric structure analysis, the following equations can be established:

$$\begin{aligned}\gamma &= \arctan\left(\sqrt{2}\tan\alpha\right)\\ h &= \sqrt{2}W_i/\tan\gamma\end{aligned} \tag{34}$$

where $\gamma$ is the interior braiding angle.

In the braiding process, the yarns are deformed due to their mutual squeezing and they are in close contact with each other. Based on the microscopic observations, the following assumptions are proposed: (1) the yarns are in completely tightened state; (2) the hexagon with the parameters $a$, $b$, $\phi$ represents the cross section of braiding yarn, as shown in Fig. 4(a); (3) the octagon with the parameter $c$ is applied for the cross section of axial yarn, as illustrated in Fig. 4(b). According to the topological structure of the RVC and the assumptions of the cross section for yarns, the following equations can be established [4]:

$$\begin{aligned}\phi &= 2\arcsin\left(\sqrt{\frac{1+2\tan^2\alpha}{2(1+\tan^2\alpha)}}\right)\\ 4&\left(a^2\sin\phi+2ab\sin\frac{\phi}{2}\right)\varepsilon = N_{f,a}\pi d_{f,a}^2\\ a &= \left(\sqrt{2}W_i-\left(3\sqrt{2}+4\right)c\right)\Big/(8\sin(\phi/2))\\ b &= \frac{W_i}{2\sin(\phi/2)\sqrt{\tan^2\alpha+1}}-\frac{\sqrt{2}W_i-\left(3\sqrt{2}+4\right)c}{4\tan(\phi/2)}\\ c &= \frac{d_{f,b}}{4}\sqrt{2\left(\sqrt{2}-1\right)N_{f,b}\pi\Big/\varepsilon}\end{aligned} \tag{35}$$

where $\varepsilon$ is the yarn packing factor and its value equals to 0.8 in the model; $N_{f,a}$ and $N_{f,b}$ are the number of filaments for the axial and braiding yarns, respectively; $d_{f,a}$ and $d_{f,b}$ are the diameters of filaments for the axial and braiding yarns with the values being equal to 7.0 μm. Based on Eq. (35), the RVC for 3D five-directional braided composites can be established, which is the consistent with experimental samples of literature, as shown in Fig. 5 (due to space limitations, only the finite element model for 40° braided composites is presented). Because of the complexity of the meso-

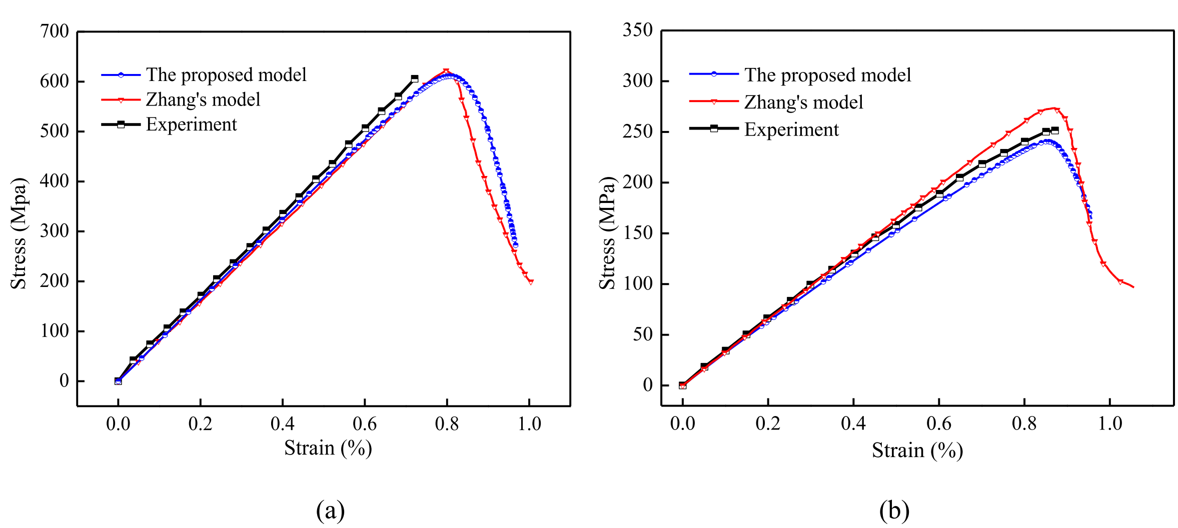


**Fig. 6.** The macroscopic stress-strain curves of 3D braided composites: (a) 20° braided composites and (b) 40° braided composites.

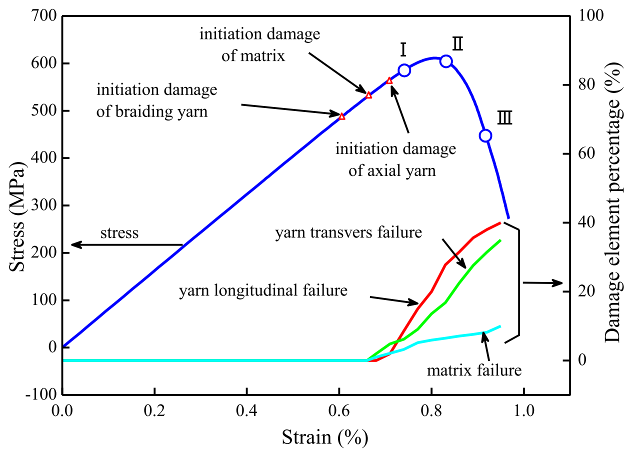


**Fig. 7.** The initial point of damage and damage element percentage for 20° braided composites.

structure, C3D4 elements are used in ABAQUS/CAE for meshing the whole model. The specimen 1 (20° braided composites) is built by 38,573 nodes and 215,032 elements, and the finite element model for the specimen 2 (40° braided composites) consists of 38,188 nodes and 212,545 elements.

### 4.3. Periodic boundary conditions

Since the elastic-plastic finite element analysis is performed for the RVC, the periodical boundary conditions (PBCs) should be applied to insure the compatibility of deformations and continuity of stresses. The mathematical expressions of PBCs for the RVC have been proposed by Xia et al. [26], which can be implemented by the multi-point constraints (MPCs) in the finite element analysis. A Python script has been developed for this purpose.

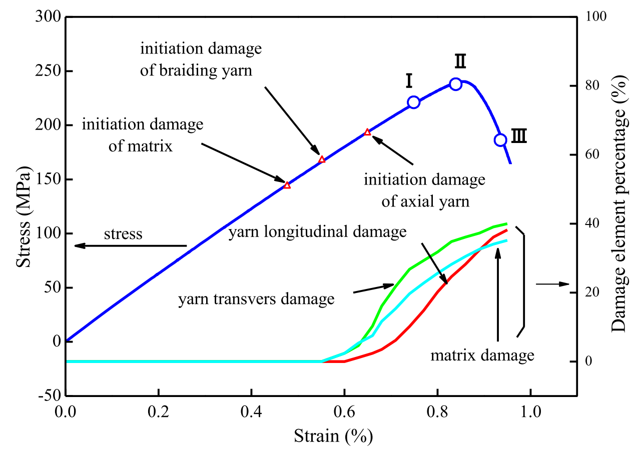


**Fig. 9.** The initial point of damage and damage element percentage for 40° braided composites.

### 4.4. Material property characterization

For braided composites, the resin matrix is assumed to be isotropic and the yarn is regarded as a unidirectional composite which has transversely isotropic elastic properties. The material

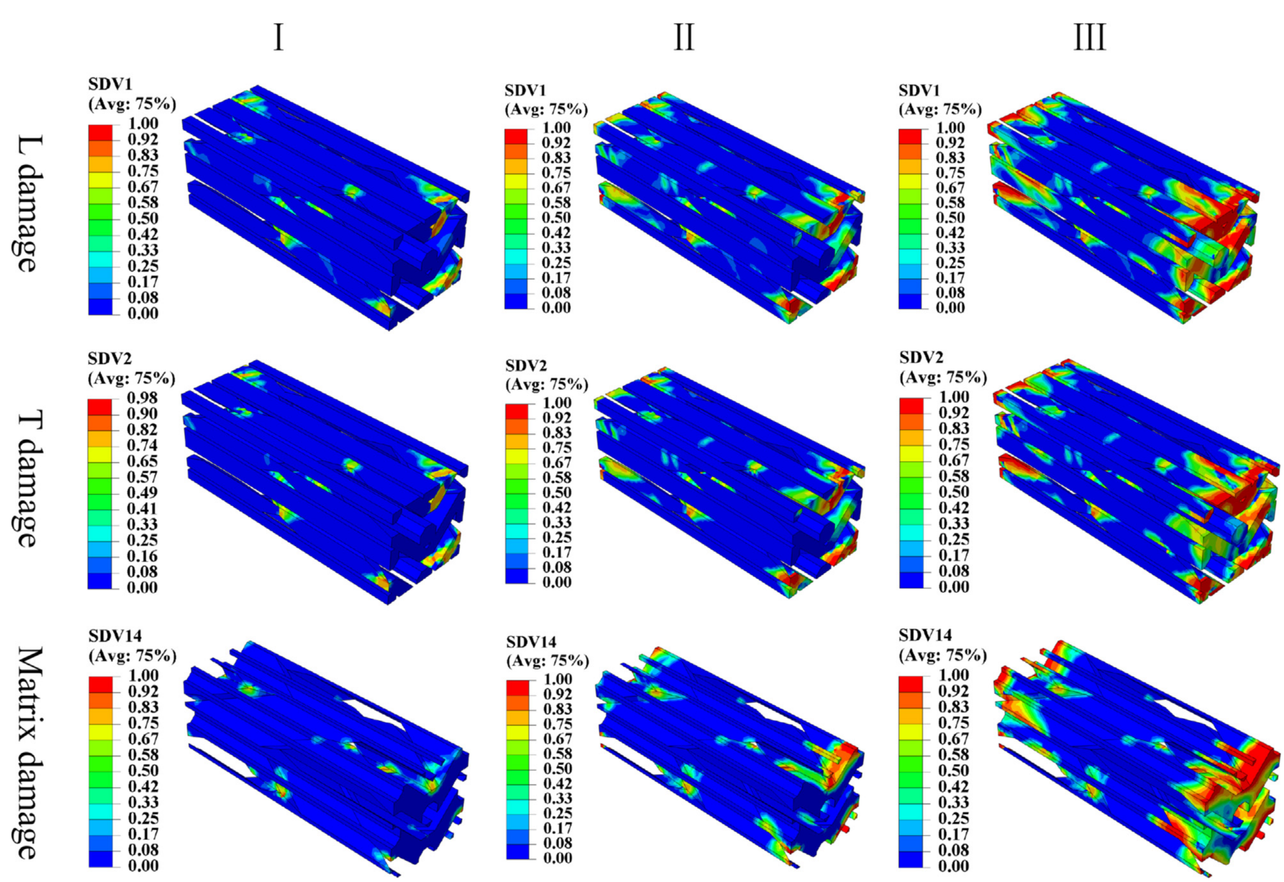


**Fig. 8.** Damage evolution process for 20° braided composites.

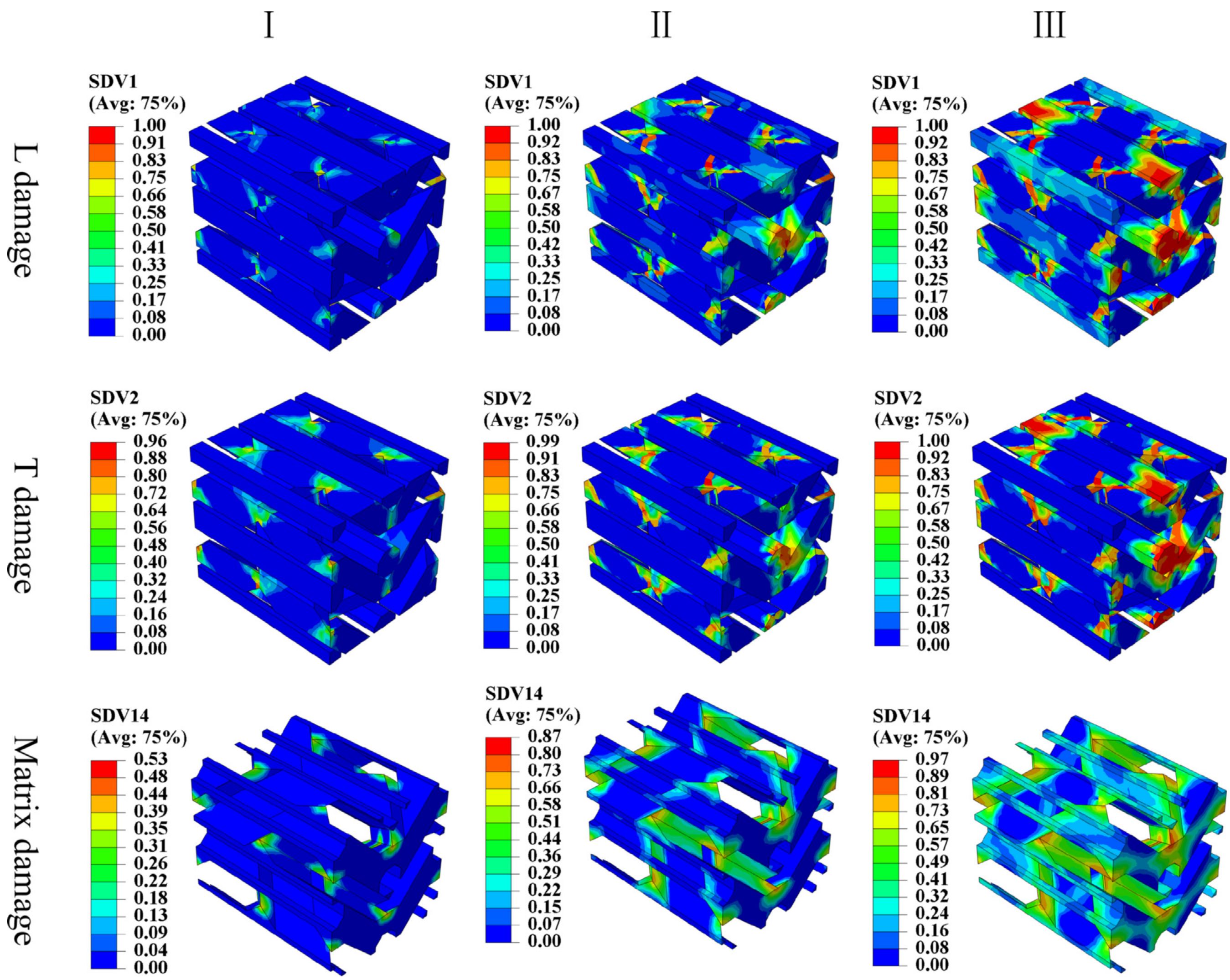


**Fig. 10.** Damage evolution process for 40° braided composites.

properties of T300 filament taken from Ref. [25] are given in Table 2, and the elastic-plastic stress-strain curve of phenolic resin obtained from Ref. [27] is applied for the finite element analysis. The material properties of the yarn including stiffness and strength can be calculated by the empirical formulae proposed by Chamis [28]. Table 3 lists the material properties of matrix and yarn for simulation, and the values of the fracture toughness are taken from Ref. [29].

## 5. Results and discussion

### 5.1. Stress-strain curve

The macroscopic stress-strain curve is obtained by the load-displacement curve at the loading point. The stress-strain curves of 20° and 40° braided composites calculated by applying the proposed model and Zhang's model [25] are compared to experimental data from Ref. [12], as shown in Fig. 6. As can be seen, for 20° braided composites, the two models give almost the same results; for 40° braided composites, the proposed elastic-plastic damage model provides much better results compared to the Zhang's model, and the results of the proposed model are very close to the experiment. The differences between two models could be explained that the Zhang's model uses linear damage evolution law to describe the stiffness degradation, and the plasticity of the matrix is not considered. While in the proposed model, the stiffness degradation is based on the exponential damage evolution law and the elastic-plastic damage of the matrix is taken into consideration.

Further analysis shows that there are some differences between the stress-strain curves of 3D braided composites of two braiding angles. For the 20° braided composites, as illustrated in Fig. 6(a), at first, the curve shows a good linearity. With continued loading, it shows a little nonlinearity. When the stress comes up to the ultimate stress, the curve decreases rapidly showing the appearance of the brittle fracture. For the 40° braided composites, as shown in Fig. 6(b), initially, the curve remains approximately linear until the strain reaches 0.6%. With continued loading, the curve shows obvious nonlinearity. Eventually, it decreases after reaching the maximum stress. The differences of macroscopic stress-strain curve between 20° and 40° braided composites can be attributed to the diverse damage and failure mechanisms which will be discussed in detail later.

### 5.2. Damage development

The initial point of damage and damage element percentage of constituent materials for 20° braided composites are shown in Fig. 7. It is noted that the damage element percentage is the ratio of damage element number to the number of constituent element of the RVC. The longitudinal and transverse damage of braiding yarns occur firstly where the yarns are squeezed together. Then, the damage of the matrix appears since the stress concentration occurs easily where the yarns and matrix contacted. Finally, the damage of axial yarns occurs with the continued longitudinal loading. Beyond the ultimate stress, the longitudinal and transverse damage element percentages of yarns reach 35-40%. At the end of the curve, the longitudinal damage element percentage is a bit more than the transverse one. However, the damage element percentage of matrix is small. Fig. 8 shows the development of the longitudinal and transverse damage of yarns and the elastic-plastic damage of the matrix to three points (I, II and III as illustrated in Fig. (7)), which can display the process more intuitively. It can be concluded that the longitudinal and transverse failure of axial yarns for 20° braided composites are the primary failure modes, which is consistent with the phenomenon observed by SEM in Refs. [3,12].

The initial point of damage and damage element percentage of constituent materials for 40° braided composites are illustrated in Fig. 9. The initial damage of 40° braided composites appears earlier than that of the 20° braided composites. The damage of matrix occurs firstly where the braiding yarns and matrix are contacted. With the continued loading, the transverse damage of braiding yarns appears because the transverse constituent materials reach the ultimate strength easily at a large braiding angle. Finally, the longitudinal and transverse damage of axial yarns occur due to the excessive tensile loading. Fig. 10 shows the developments of the longitudinal and transverse damage of yarns and the elastic-plastic damage of the matrix to three points (I, II and III as sketched in Fig. 9). Obviously, the damage element percentage of the matrix for the 40° braided composites is much more than that of the 20° braided composites. The elastic-plastic damage of the matrix estimated by the proposed model is observed to be coincident with the SEM graphs of the 40° braided composites in Refs. [3,12]. This analysis leads to the conclusion that the stress-strain curve of the 40° braided composites is controlled by the longitudinal failure of axial yarns, transverse failure of braiding yarns and elastic-plastic failure of matrix.

As the conclusion, the proposed model could provide accurate and reliable simulations for the stress-strain curve and reflect the damage and failure mechanisms of the 3D five-directional braided composites. Moreover, the model has offered a method to take into account the elastic-plastic damage of matrix.

## 6. Conclusions

This paper has discussed a novel damage model based on the framework of Continuum Damage Mechanics for the description of damage accumulation of 3D braided composites. The proposed model predicts the onset and propagation of fiber bundle using an elastic damage model, and it describes the degradation of matrix applying an elastic-plastic damage model. The damage evolution in each failure mode of the yarn bundle and matrix is controlled by a different internal variable. The Bazant crack band theory is adopted to ensure a mesh-independent estimation of energy dissipation for numerical implementation. The new proposed model is examined by the quasi-static uniaxial tests of 3D braided composites from the literature, and the numerical results correlate well with the experiment. It seems to support the idea that the proposed damage method is very useful for the evaluation of damage mechanisms for 3D braided composites.

However, further experiment for the 3D braided composites under quasi-static tensile and cyclic loadings would be needed for an advanced validation of the proposed elastic-plastic damage model, and the effects of temperature and interface will be taken into account to complete the model.

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## Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.compscitech.2018.01.027.

## References

[1] A.P. Mouritz, M.K. Bannister, P.J. Falzon, K.H. Leong, Review of applications for advanced three-dimensional fibre textile composites, Compos. Part A-Appl. S 30 (12) (1999) 1445–1461.
[2] K. Bilisik, Three-dimensional braiding for composites: a review, Textil. Res. J. 83 (13) (2013) 1414–1436.
[3] D.S. Li, Z.X. Lu, L. Chen, J.L. Li, Microstructure and mechanical properties of three-dimensional five-directional braided composites, Int. J. Solid Struct. 46 (18) (2009) 3422–3432.
[4] D.S. Li, D.N. Fang, N. Jiang, X.F. Yao, Finite element modeling of mechanical properties of 3D five-directional rectangular braided composites, Compos. B Eng. 42 (6) (2011) 1373–1385.
[5] X.G. Yu, J.Z. Cui, The prediction on mechanical properties of 4-step braided composites via two-scale method, Compos. Sci. Technol. 67 (3) (2007) 471–480.
[6] V. Carvelli, C. Poggi, A homogenization procedure for the numerical analysis of woven fabric composites, Compos. Part A-Appl. S 32 (10) (2001) 1425–1432.
[7] G.D. Fang, J. Liang, Y. Wang, B.L. Wang, The effect of yarn distortion on the mechanical properties of 3D four-directional braided composites, Compos. Part A-Appl. S 40 (4) (2009) 343–350.
[8] C. McGregor, N. Zobeiry, R. Vaziri, A. Poursartip, X. Xiao, Calibration and validation of a continuum damage mechanics model in aid of axial crush simulation of braided composite tubes, Compos. Part A-Appl. S 95 (2017) 208–219.
[9] G.D. Fang, J. Liang, B.L. Wang, Progressive damage and nonlinear analysis of 3D four-directional braided composites under unidirectional tension, Compos. Struct. 89 (1) (2009) 126–133.
[10] T. Wehrkamp-Richter, R. Hinterhölzl, S.T. Pinho, Damage and failure of triaxial braided composites under multi-axial stress states, Compos. Sci. Technol. 150 (2017) 32–44.
[11] D.S. Ivanov, F. Baudry, B. Van Den Broucke, S.V. Lomov, H. Xie, I. Verpoest, Failure analysis of triaxial braided composite, Compos. Sci. Technol. 69 (9) (2009) 1372–1380.
[12] Z.P. Li, Z.X. Lu, Z.H. Feng, D.S. Li, Investigation of the tensile properties and failure mechanism of integrally-braided 5D carbon/phenolic composites, Acta Aeronaut. Astronaut. Sin 28 (4) (2007) 869–873.
[13] G.D. Fang, J. Liang, Q. Lu, B.L. Wang, Y. Wang, Investigation on the compressive properties of the three dimensional four-directional braided composites, Compos. Struct. 93 (2) (2011) 392–405.
[14] S. Song, A.M. Waas, K.W. Shahwan, X. Xiao, O. Faruque, Braided textile composites under compressive loads: modeling the response, strength and degradation, Compos. Sci. Technol. 67 (15) (2007) 3059–3070.
[15] A. Matzenmiller, J. Lubliner, R.L. Taylor, A constitutive model for anisotropic damage in fiber-composites, Mech. Materials 20 (2) (1995) 125–152.
[16] J. Lemaitre, R. Desmorat, Engineering Damage Mechanics: Ductile, Creep, Fatigue and Brittle Failures, Springer, Berlin, 2005.
[17] Z. Hashin, Failure criteria for unidirectional fiber composite, J. Appl. Mech. 47 (1980) 329–334.
[18] R. Faria, J. Oliver, M. Cervera, A strain-based plastic viscous-damage model for massive concrete structures, Int. J. Solid Struct. 35 (14) (1998) 1533–1558.
[19] Z.P. Bazant, J. Planas, Fracture and Size Effect in Concrete and Other Quasi-brittle Materials, CRC Press, Boca Raton, 1998.
[20] P. Maimí, P.P. Camanho, J.A. Mayugo, C.G. Dávila, A continuum damage model for composite laminates: Part II–Computational implementation and validation, Mech. Mater. 39 (10) (2007) 909–919.
[21] Z.P. Bažant, B.H. Oh, Crack band theory for fracture of concrete, Mater. Struct. 16 (3) (1983) 155–177.
[22] Simulia, ABAQUS 6.13 User's Manual. Providence, 2013. Rhode Island, US.
[23] J.C. Simo, T.J. Hughes, Computational Inelasticity, Springer, New York, 1998.
[24] G. Duvant, J.L. Lions, Inequalities in Mechanics and Physics, Springer, Berlin, 1976.

[25] C. Zhang, J.L. Curiel-Sosa, T.Q. Bui, A novel interface constitutive model for prediction of stiffness and strength in 3D braided composites, Compos. Struct. 163 (2017) 32–43.
[26] Z.H. Xia, C.W. Zhou, Q.L. Yong, X.W. Wang, On selection of repeated unit cell model and application of unified periodic boundary conditions in micro-mechanical analysis of composites, Int. J. Solid Struct. 43 (2) (2006) 266–278.
[27] A. Gardziella, L.A. Pilato, A. Knop, Phenolic Resins: Chemistry, Applications, Standardization, Safety and Ecology, second ed., Springer, Berlin, 2000.
[28] C.C. Chamis, Mechanics of composite materials: past, present, and future, J. Compos. Technol. Res. 11 (1) (1989) 3–14.
[29] X. Li, W.K. Binienda, R.K. Goldberg, Finite-element model for failure study of two-dimensional triaxially braided composite, J. Aero. Eng. 24 (2) (2010) 170–180.